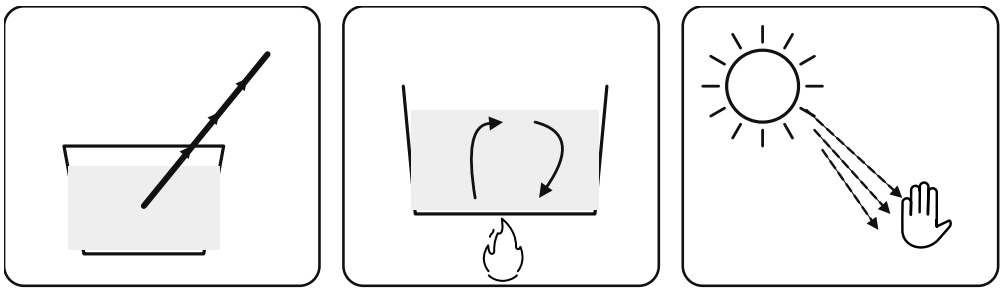
PHYSICAL AND CHEMICAL SCIENCE

Heat Transfer: Conduction, Convection and Radiation

How heat moves through solids, liquids, gases and empty space, and how we control it with conductors, insulators and colours



conduction
heat travels through solids

convection
warm liquid or air rises

radiation
heat travels as rays

In this pack

- **Learn it:** How heat travels ★☆☆
- **Fact check:** Conduction, convection or radiation? ★☆☆
- **Think it through:** Explaining heat in daily life ★★☆
- **Investigate:** Which colour heats up fastest in the sun? ★★☆
- **Real-world case study:** Keeping cool in a hotter Dhaka ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C9 · HEAT TRANSFER

How heat travels

★☆☆
Grade 5

Name: Class: Date:

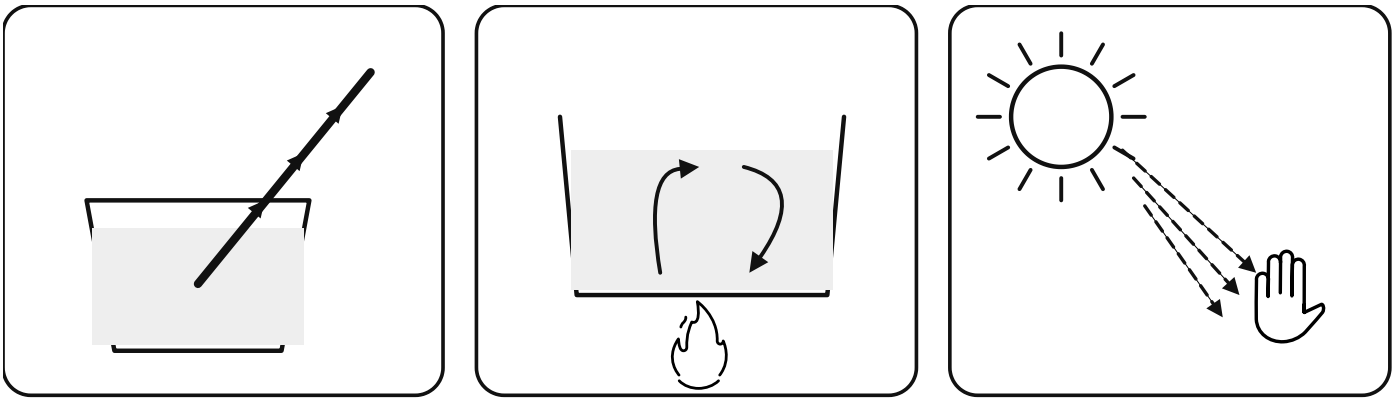
Learning goals

- Explain the difference between heat and temperature.
- Describe conduction, convection and radiation using the particle model.
- Explain how conductors, insulators and colours are used to control heat.

Hold a cup of hot tea and your hands warm up. Stand in the sun and your face feels hot. Heat is always on the move, from **hotter** things to **colder** things, until they reach the same temperature.

Heat and temperature

Heat is energy that moves because of a difference in temperature. **Temperature** (in °C) tells us how hot something is: how fast its particles are moving. A bucket of warm water holds more heat than a cup of boiling water, even though the cup is hotter.



conduction
heat travels through solids

convection
warm liquid or air rises

radiation
heat travels as rays

The three ways heat travels.

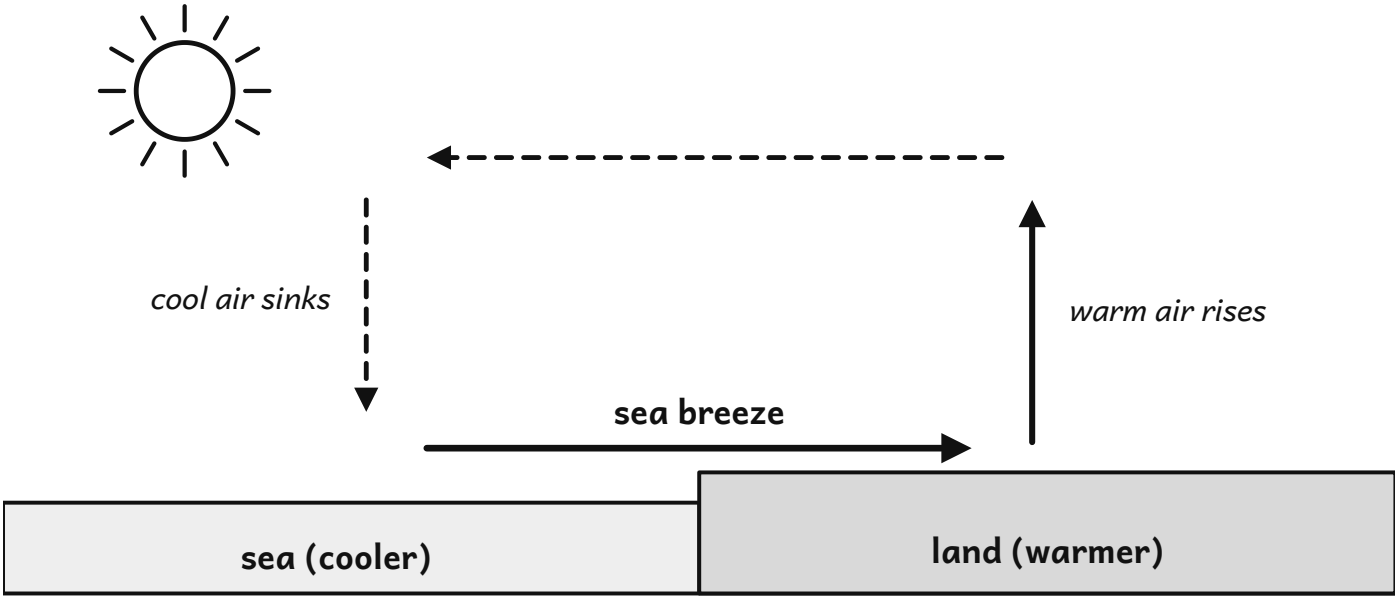
1. Conduction

In a **solid**, heat passes from particle to particle. The hot particles vibrate faster and bump into their neighbours, passing the energy along. A metal spoon in hot tea soon feels hot at the handle.

Conductors let heat through easily: metals such as copper, aluminium and steel. **Insulators** do not: wood, plastic, cloth, wool and especially **trapped air**.

2. Convection

In **liquids and gases**, warm parts **expand**, become less dense and **rise**. Cooler, denser parts **sink** to take their place. This loop is a **convection current**. It stirs a pot of water on the stove and makes the daytime **sea breeze**.

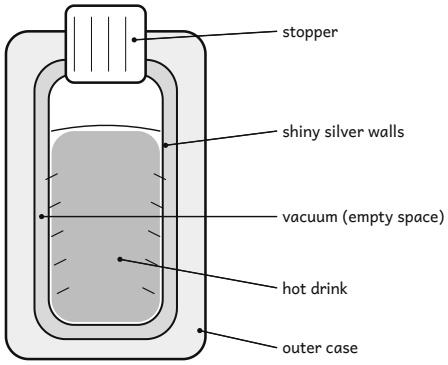


A sea breeze is a convection current.

3. Radiation

Heat can travel as invisible **infrared rays**, even through empty space. That is how the Sun's energy reaches Earth. **Dark, dull** surfaces absorb radiation well and heat up. **Light, shiny** surfaces reflect it.

Putting it together: a vacuum flask



A vacuum flask slows all three kinds of heat transfer.

The **vacuum** has no particles, so no conduction or convection can cross it. The **shiny walls** reflect radiation. The **stopper** stops warm air rising out by convection. The same flask keeps cold drinks cold!

Bangladesh connection

Traditional homes in Bangladesh use these ideas: thick mud walls and thatched roofs are good insulators, high ceilings let hot air rise away, and trees shade roofs from radiation.

Key words

conduction	heat passing through a solid, particle to particle
convection	heat carried by moving liquid or gas
radiation	heat travelling as infrared rays
conductor	lets heat through easily
insulator	slows the flow of heat
convection current	a loop of rising warm and sinking cool fluid

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FACT CHECK · C9 · HEAT TRANSFER

Conduction, convection or radiation?

★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Identify the type of heat transfer.
- Classify conductors and insulators.

1 Which kind of heat transfer is it? [6]

Example	Conduction, convection or radiation?
A metal pan handle gets hot.	
Warm air rises from a heater.	
The Sun warms your face.	
Your feet feel cold on a tiled floor.	
A hot-air balloon rises.	
You feel warm in front of a fire.	

2 Sort these materials. [6]

SORT THESE

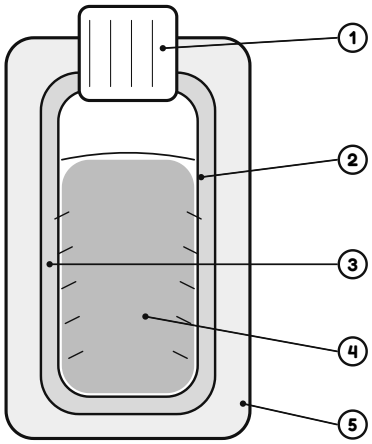
steel copper wool trapped air aluminium wood

Good conductor	Good insulator

3 **Label the vacuum flask.** [5]

LABELS

stopper **shiny silver walls** **hot drink** **vacuum (empty space)** **outer case**



1 _____ 2 _____

3 _____ 4 _____

5 _____

THINK IT THROUGH · C9 · HEAT TRANSFER

Explaining heat in daily life

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Explain with the particle model.
- Apply ideas to design.

1 **Explain, using particles, why a metal spoon in hot dal gets hot all the way up.** [3]

2 **Why are heaters put near the floor but air conditioners high on the wall?** [3]

3 **Why do people in hot countries often wear white or light-coloured clothes?** [2]

4 **A cook's pan has a copper base and a plastic handle. Explain both choices.** [2]

Which colour heats up fastest in the sun?

★★☆ Grade 5

Name: Class: Date:

Learning goals

- Plan a fair test.
- Plot and interpret a line graph.

Class 5B wrapped three identical cans of water in black paper, white paper and shiny foil. They put them in the sun and measured the water temperature every 10 minutes.

1 **Name the variables.** [3]

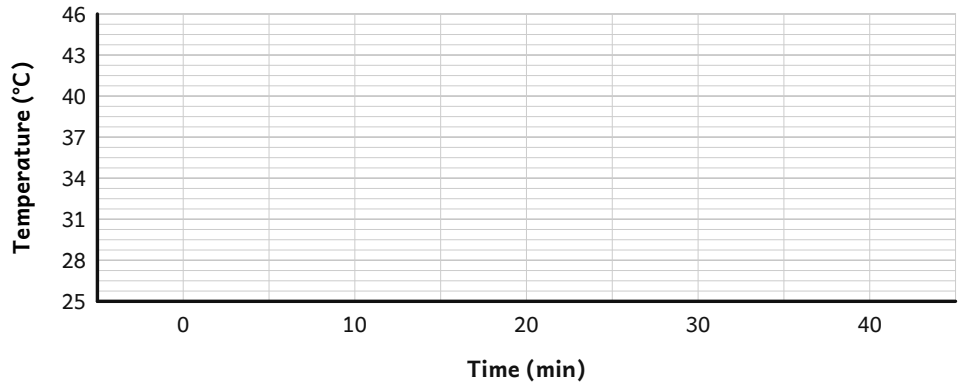
a) Changed:

b) Measured:

c) Kept the same:

2 **Plot the three lines on one graph (use a key).** [4]

Time (min)	0	10	20	30	40
black (°C)	28	33	38	42	45
white (°C)	28	30	32	34	35
shiny foil (°C)	28	29	30	31	32



3 **Which heated fastest? Explain why using the word radiation.** [2]

4 **Use your results to choose *a* colour for *a* water tank on *a* roof that should stay cool.** [2]

.....

.....

Data: example results for practice.

REAL-WORLD CASE STUDY · C9 · HEAT TRANSFER

Keeping cool in a hotter Dhaka

★★★ Grade 5

Name: Class: Date:

Learning goals

- Apply heat transfer to a real problem.
- Compare solutions using data.

Science | Design and technology | Health | Maths

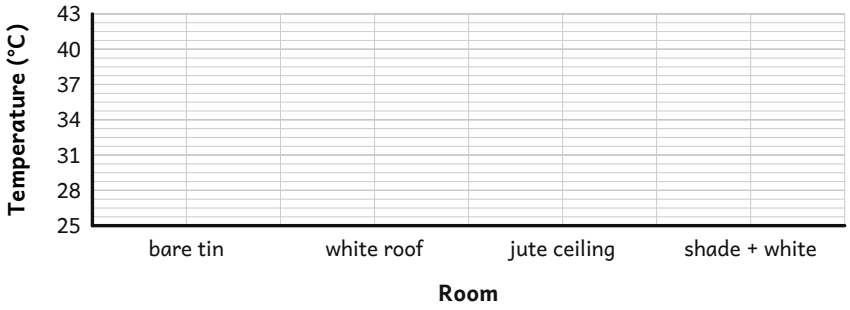
Many homes in Dhaka have thin **tin (corrugated iron) roofs**. On a sunny day the dark metal absorbs radiation and conducts heat straight into the room below. Inside, it can be much hotter than outside, which is dangerous for babies and older people during a heatwave.

Some families have tried cheap ideas: painting the roof **white**, adding a **ceiling of jute or bamboo** with an air gap, and growing **creepers or trees** for shade. A community group measured the temperature inside four rooms at 2 pm on the same hot day.

Example data based on cool-roof trials in hot cities; numbers are illustrative.

1 **Draw a bar chart of the results.** [3]

Room	bare tin roof	white-painted roof	jute ceiling with air gap	tree shade + white roof
Temperature at 2 pm (°C)	41	36	35	32



2 **Explain why the white roof was cooler, using radiation.** [2]

.....

.....

3 **Explain why the jute ceiling with an air gap helped, using conduction.** [2]

.....

.....

4 **Recommend one solution for a poor family. Give a scientific and a money reason.** [3]

.....

.....

Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the words.**

S	O	L	K	A	Y	W	C	H	I	T
R	A	D	I	A	T	I	O	N	I	T
I	I	J	U	T	V	L	N	S	H	H
N	A	H	Y	K	H	K	V	N	G	J
S	M	H	E	A	T	U	E	I	H	W
U	V	A	C	U	U	M	C	A	K	K
L	C	O	N	D	U	C	T	I	O	N
A	N	H	J	W	P	U	I	I	R	U
T	M	E	T	A	L	P	O	J	A	H
O	F	C	B	L	G	M	N	Y	W	V
R	J	V	P	H	H	C	L	S	J	D

conduction **convection** **heat** **insulator** **metal** **radiation** **vacuum**

2 **Heat travels through empty space by...** [1]

- ◯ conduction
- ◯ convection
- ◯ radiation

3 **Which is the best insulator?** [1]

- ◯ copper
- ◯ steel
- ◯ wool

4 **Challenge. Design a lunch box that keeps rice hot for 4 hours. Label how it reduces conduction, convection and radiation.** [4]

My hot lunch box

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain heat and temperature	○	○	○
describe conduction, convection and radiation	○	○	○
choose conductors and insulators	○	○	○
plot a line graph of heating	○	○	○
design ways to keep cool	○	○	○